

PARKINSON DISEASE DETECTION

MINOR PROJECT REPORT

Submitted in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

(Department of Computer Science and Engineering)

Submitted to

**INDIAN INSTITUTE OF INFORMATION TECHNOLOGY
BHOPAL (M.P.)**



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CERTIFICATE

This is to certify that the work embodied in this report entitled “**PARKINSON DISEASE DETECTION**” has been satisfactorily completed by **SHIVANSHU YADAV(23U02036)** and **ABHINAV PRAJAPATI(23U02074)**. It is a bonafide piece of work, carried out under our guidance in the **Department of Computer Science and Engineering, Indian Institute of Information Technology, Bhopal** for the partial fulfillment of the Bachelor of Engineering during the academic year 2025-26.

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DECLARATION

We hereby declare that the following minor project synopsis entitled “PARKINSON DISEASE DETECTION” presented in the is the partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in** Computer Science and Engineering. It is an authentic documentation of our original work carried out under the able guidance of **Dr. Neeta Anna Eapen**. The work has been carried out entirely at the Indian Institute of Information Technology, Bhopal. The project work presented has not been submitted in part or whole to award of any degree or professional diploma in any other institute or organization.

We, with this, declare that the facts mentioned above are true to the best of our knowledge. In case of any unlikely discrepancy that may occur, we will be the ones to take responsibility.

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ACKNOWLEDGEMENT

We, Shivanshu Yadav and Abhinav Prajapati, would like to express our sincere gratitude to everyone who supported and guided us throughout the successful completion of our minor project titled "PARKINSON DISEASE DETECTION".

First and foremost, we would like to extend our heartfelt thanks to our esteemed **Director**, Dr. Ashutosh Kumar Singh, for providing us with the necessary resources and facilities for conducting our project. His encouragement and vision played a key role in helping us accomplish our goals.

We are deeply grateful to our **Project Guide**, Dr. Neeta Anna Eapen, for her constant guidance, invaluable suggestions, and expert advice throughout the project. Her knowledge and insights helped us enhance the quality of our work, and her support kept us motivated during difficult times.

Our sincere thanks to the **Project Coordinator**, Dr. Rahul Chaurasia, for organizing and overseeing the entire project process. His/her assistance in clarifying doubts and providing us with the right direction was instrumental in the smooth execution of our project.

We would also like to express our gratitude to all the **faculties** of the Computer Science Engineering for their continuous support and encouragement. Their teaching and knowledge were invaluable in enhancing our technical skills and understanding, which helped us in the successful completion of the project.

A special thanks to our **friends** for their continuous support, valuable suggestions, and helping us during various stages of the project. Their insights and encouragement motivated us to keep working on the project with enthusiasm.

Last but certainly not the least, we would like to express our deepest gratitude to our **parents** and family members for their love, patience, and continuous moral support throughout the course of this project. Their understanding and encouragement were the driving force behind our efforts.

We are truly thankful to everyone who directly or indirectly contributed to the success of this project. Without their assistance and support, this project would not have been possible.

Shivanshu Yadav (23U02036)
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AREA OF WORK

Our project primarily focuses on HEALTHCARE ANALYTICS and MACHINE LEARNING, utilizing advanced algorithms and data processing techniques essential for developing intelligent diagnostic systems.

The detection and early diagnosis of Parkinson's disease represent a critical challenge in modern healthcare. Traditional diagnostic methods often rely on subjective assessments and may not detect the disease in its early stages. This is where machine learning comes into play, offering objective, data-driven approaches to disease detection.

In recent years, the application of machine learning in medical diagnostics has shown tremendous growth. The ability to analyse complex patterns in medical data, including voice recordings, motor function tests, and clinical assessments, has made it possible to develop automated systems that can assist healthcare professionals in making more accurate and timely diagnoses.

In our project, we explore various machine learning algorithms including Support Vector Machines, Random Forest, Neural Networks, and ensemble methods to build a robust classification model. We analyse biomedical voice measurements and other clinical features to predict the presence of Parkinson's disease with high accuracy.

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ABSTRACT

Parkinson's disease (PD) is a progressive neurodegenerative disorder characterized by the deterioration of motor functions due to dopaminergic neuron loss in the *substantia nigra*. Early diagnosis is critical for initiating neuroprotective interventions; however, traditional diagnostic methods rely heavily on subjective clinical assessments of overt motor symptoms, which often manifest only after significant neurological damage has occurred. Recent clinical research has identified hypokinetic dysarthria—a complex speech impairment manifesting as vocal perturbation—as one of the earliest prodromal biomarkers of PD. This research proposes an objective, non-invasive, and highly accessible machine learning framework designed to detect Parkinson's disease utilizing digital acoustic voice analysis.

The methodology utilized the "Parkinsons Disease Dataset" sourced from the UCI Machine Learning Repository, comprising 195 instances and 22 distinct acoustic biomedical features (e.g., Jitter, Shimmer, HNR, and non-linear dynamic metrics like PPE and DFA) extracted from sustained vowel phonations. Exploratory data analysis revealed a severe class imbalance (147 PD instances vs. 48 healthy controls) and significant multicollinearity among feature subsets. To ensure algorithmic stability and prevent majority-class bias, the Synthetic Minority Over-sampling Technique (SMOTE) was employed to synthesize minority instances, followed by Min-Max feature scaling to normalize the multidimensional data into a standardized space $[-1, 1]$.

INTRODUCTION

Parkinson's Disease (PD) is a neurodegenerative condition that gradually and progressively impairs the functioning of the central nervous system, primarily the motor system. From a biological perspective, PD is an ailment where there is a gradual loss and eventual death of dopaminergic neurons in the substantia nigra, which is a vital part of the midbrain responsible for reward behaviour and movement control. The neurotransmitter dopamine is crucial in facilitating the transmission of impulses from the substantia nigra to the corpus striatum to ensure the smooth execution of intentional muscle movements. Loss of around 60% to 80% of the neurons in the substantia nigra can affect the brain's capacity to coordinate voluntary movements.

The main symptomatology of Parkinson's disease is associated with the presence of four motor symptoms that are described by an acronym, TRAP, which stands for Tremor at rest (most often occurs on one side of the body initially), Rigidity (muscle stiffness), Akinesia/Bradykinesia (slow movement initiation and performance), and Postural instability (causing difficulties balancing and often resulting in falls). Apart from the aforementioned motor disturbances, a variety of other non-motor symptoms can also be seen, such as autonomic problems, sleep disturbance, cognitive impairment, depression, and altered sensation, including anosmia. [1]

In terms of epidemiology, Parkinson's disease can be stated to be the second most prevalent neurodegenerative disorder globally, with only Alzheimer's disease being more prevalent than Parkinson's disease. As reported by the Parkinson's Foundation and WHO, there are about 10 million people suffering from PD worldwide. The incidence of the disease has been found to be very strongly linked to increasing age, and the most affected population is that which is above 60 years old. Nevertheless, early onset PD, which affects persons below the age of 50, is estimated to occur in 10% of all PD cases.

Background

In the past, the diagnosis of Parkinson's Disease has been primarily based on subjective assessment and patient history. Neurologists usually employ well-defined rating scales for clinical examination, such as the Unified Parkinson's Disease Rating Scale (UPDRS), which measures the extent of motor symptom severity. Although these clinical assessments have remained the benchmark until now, they are highly problematic. For one, they involve subjective judgment and are largely dependent upon the expertise and observational skills of the attending physician. Furthermore, by the time patients exhibit clinical signs of the disease, they have incurred significant neurodegeneration. In short, there is a clinical necessity to detect the disease in the "prodromal" phase—when it manifests itself but the classical symptoms are not yet observable.[2]

In the last few years, ML/AI has brought about a paradigm shift in the field of diagnostic technology. Unlike the previous statistical approach, The ML algorithms inherently have the ability to process large amounts of intricate data and detect patterns which escape human

detection through their non-linearity. Through training such algorithms on data sets of quantified measurements of biomedical voices, it is possible to create highly accurate and automated systems for screening people. Such systems would enable us to use advanced technology to analyze subtle variations in our vocal acoustics (like jitter, shimmer, etc.), making it possible to differentiate between patients and healthy people suffering from the early symptoms of vocal changes caused by Parkinsonism.

Motivation

The main driver behind this research is the immediate need worldwide to develop a diagnostic test that is affordable, non-invasive, and very inexpensive. Conventional tests may include costly neuroimaging procedures like DaTscans, invasive lumbar punctures to examine the cerebrospinal fluid, or prolonged periods waiting to consult movement disorder specialists. In most cases, such factors hinder timely and precise diagnoses, especially among people living in remote areas or economically deprived communities.[3]

Utilizing the human voice as a diagnostic marker is an incredibly promising and revolutionary method because of the following reasons:

- The Lack of Need for Any Specific Equipment:** While an MRI machine, CT scan, or even certain laboratory tests may be required for the diagnosis, collecting the voice data requires no other specialized device but a regular microphone, which is present in all smartphones, tablets, and computers.
- The Possibility to Deliver Objective Metrics:** The metrics obtained from machine learning models are based on mathematical calculations and are free from human subjectivity present in standard observation techniques. It establishes a common metric for patients' evaluation.
- The Capacity to Act Early:** As vocal deterioration tends to occur before any limb problems appear, the use of highly accurate voice analysis technology helps identify the patients at risk of developing neurodegenerative diseases years before their first symptoms appear.
- The Opportunity to Conduct Continuous Monitoring:** The collection of voice data makes it easy to monitor the progress of the disease and the effects of medications long term. For example, one could undergo a 30-second-long voice test daily, giving clinicians access to a detailed map of one's condition changes throughout the year.

Scope of the Project

This project aims at the holistic process of developing a machine learning based diagnostic system for diagnosing Parkinson's disease based only on biometric recordings of the patient's voice. This scope involves developing a complete data science workflow that enables a connection between the data analysis and clinical usage.

In particular, the project scope involves the following key steps:

1. **Collecting and Combining Data:** Using valid, publicly accessible or clinical datasets that contain high-quality voice recordings from both normal subjects and clinically confirmed patients suffering from PD. These datasets will include recordings of sustained vowels, reading exercises, and natural speech.
2. **Fetching Acoustic Features:** Using sophisticated digital signal processing technology to derive appropriate acoustic features for the voice. This involves measuring features such as vocal fold vibrations or irregularities (Jitter), variations in amplitude (Shimmer), harmonic-noise ratio (HNR), DFA, and MFCCs.
3. **Pre-processing and Reducing Dimensions:** Ensuring the consistency of data by cleaning missing values, scaling the features appropriately to achieve algorithmic stability, and reducing dimensions using statistical techniques such as PCA. Dimensionality reduction serves the purpose of removing unnecessary features and avoiding overfitting.
4. **Selecting Algorithms and Developing Models:** Developing models by training and testing a set of diverse classifiers. In this project, we will be experimenting with different types of classifiers ranging from classical models such as SVM, KNN, and logistic regression to more modern ensemble methods including random forests and gradient boosting machines.
5. **Evaluation and Validation:** Utilization of rigorous cross-validation processes (such as K-fold Cross Validation) to confirm the validity of the model when applied to new datasets. Each model's performance will be assessed in detail through metrics such as Classification Accuracy, Precision, Sensitivity (Recall), F1-Score, and the Area Under the Receiver Operating Characteristic Curve.
6. **Intepretability and Translatability:** An important limitation to consider regarding the scope of the study is maintaining transparency of the developed model, avoiding its characterization as a 'black box.' The study will make use of Explainable AI (XAI) techniques in order to understand which particular aspects of vocalizations have a strong impact on the model's decision making process.

LITERATURE REVIEW

The convergence of neurology, digital signal processing, and artificial intelligence has created an abundant and constantly growing amount of scientific literature devoted to the objective detection and tracking of PD. Conventionally, the diagnosis and monitoring of PD has been based on patient assessments using various clinical scales, for example, the Unified Parkinson's Disease Rating Scale (UPDRS); although these scales are crucial, they are necessarily subjective, episodic, and require the manifestation of motor signs. Thus, the past couple of decades have seen an interdisciplinary effort aimed at finding non-invasive and objective markers that could identify PD in its preclinical state. In this regard, voice and speech analysis stand out as the most promising approaches. The current review discusses the current state of research, focusing on acoustic features of PD, developments in machine learning techniques used for their identification, and key barriers to implementing them in clinical practice.[4]

Voice Features and Acoustic Biomarkers in Parkinson's Disease

The biological basis behind the study of PD voices stems from the profound effect the disorder has on the complete speech chain. Speaking involves the precise coordination of the breathing, phonation, and articulation chains. The signature dopaminergic insufficiency seen in PD triggers bradykinesia (slow movements), rigidity, and tremors in the muscles that control all these components, producing a multifaceted speech disorder known as hypokinetic dysarthria. This condition occurs in up to 90% of PD patients and precedes motor impairments in limbs.[5]

These impairments have been thoroughly investigated in numerous studies spanning many decades. These studies have progressed from perceptual evaluations to highly detailed digital acoustic measures. These measures can be categorized into multiple groups within the literature:

Classical Measures of Voice Perturbations

Initially, much work in computing was centered around sustained vowel sounds (asking a subject to produce the "ah" sound for as long as possible).

- **Regulation Problems With Fundamental Frequency (F0):** In normal conditions, humans are able to regulate their voice pitch during speech. However, PD causes rigidity in the movement of larynx muscles, leading to insufficient variability of fundamental frequency and producing monopitch voice.
- **Jitter (Frequency Instability):** This parameter measures micro-fluctuations of fundamental frequency from one cycle to another. It is evident from research that patients with PD have increased jitter due to poor neuro-muscular regulation of larynx muscles causing instability of voice tone.
- **Shimmer (Amplitude Instability):** Like jitter, shimmer parameter evaluates fluctuations in sound loudness in cycles. Patients with PD exhibit high values of

shimmer since PD causes poor breathing and inadequate glottal closure, making speech sound weaker and breathier. Spectral and Cepstral Features

With the increase in computing power, research moved on from just analyzing time-domain perturbations in the analysis of the frequency spectrum of the voice.

- **Harmonic-to-Noise Ratio:** HNR is the measurement of the ratio of the harmonic acoustic energy present to aperiodic or noisy acoustic energy. Since most Parkinsonian disease patients usually have vocal fold bowing and incomplete closure, the extra air results in turbulence, which creates noisy energy. As such, PD patients tend to show much lower HNR than non-PD subjects.
- **Mel-Frequency Cepstral Coefficients:** Adapted from automatic speech recognition techniques, MFCCs prove extremely useful in modeling the human vocal tract. Previous studies indicate that up to the first 13 to 40 MFCCs can describe the articulation deficiencies of PD speech, including the poor pronunciation of consonants and limited jaw-tongue mobility.

Non-Linear Dynamical Systems (NDS) Features

The advent of new non-linear algorithms was the next paradigm shift that aimed at reflecting the complexity and chaos of speech. In a breakthrough paper, Little et al. (2008) demonstrated that linear features are incapable of capturing the extremely turbulent behavior of a pathological voice.

- Little et al. introduced new features, such as Pitch Period Entropy (PPE), which quantifies the inability of PD patients to deliver constant pitch in connected speech but disregards normal pitch turbulence in healthy people.
- Different non-linear properties, like Detrended Fluctuation Analysis (DFA) and Correlation Dimension (D2), estimate the fractal scaling and complexity of an audio signal. These features are nowadays widely used in PD datasets since they exhibit high robustness to the differences in recording conditions and microphones.

Machine Learning Approaches

Feature extraction has resulted in the development of advanced classification methods to distinguish between normal and abnormal voice signals. The literature demonstrates a definite trend toward the evolution of the techniques from statistical learners to advanced ensembles and deep learning frameworks.[6]

Traditional and Kernel Classifiers

Studies conducted during the earlier stages of voice pathology detection used traditional machine learning techniques such as Naive Bayes, k-nearest neighbors (k-NN), and logistic regression. These methods began to pose challenges when dealing with large numbers of

acoustic features since there is usually more noise than signal within the dataset. This phenomenon was known as the curse of dimensionality.

Support Vector Machines (SVM)

Support vector machines (SVM) rapidly gained popularity for their application in voice analysis. Through the kernel trick (usually the radial basis function or RBF kernel), support vector machines are able to project high-dimensional acoustic data to a space that is even higher dimensional to obtain the hyperplane which best separates the control group from the patients with PD. A number of investigations have shown classification accuracies ranging between 85% and 92%.

Ensemble Methods

In order to increase the accuracy and robustness of algorithms, the use of ensemble learning methodologies was introduced.

Ensemble learning methodology uses multiple base learners in order to generate more accurate results. The well-known paper by Sakar et al. (2013) illustrated the strength of the ensemble approach. Through the use of ensembles, they managed to reach accuracies higher than 95%.

Feature Engineering and Selection Optimization

It is also evident that increasing the volume of input data to the algorithm is not always beneficial because excessive acoustic features may have a detrimental impact on algorithm accuracy. Thus, feature engineering and feature selection become essential steps in an ML workflow.

Dimensionality Reduction Methods

Acoustic extraction tools like Praat and openSMILE are capable of extracting thousands of low-level audio features from just two seconds of speech recording. This is why scholars make use of methods such as Principal Component Analysis (PCA). The purpose of using PCA is to convert the high-dimensional space of acoustic features into a set of linearly uncorrelated principal components on which the machine learning model is trained.[7]

Feature Selection Methods

It becomes clear from research literature that not all voice parameters are equally significant for diagnosis. One of the important challenges faced by scientists is identifying the best "feature subset."

- **Filter Methods:** Techniques such as Pearson correlation, ANOVA, and Chi-square tests are used to filter out features that have little to no correlation with the target variable (PD vs. Healthy).

- **Wrapper Methods:** Recursive Feature Elimination (RFE) is widely documented in PD literature. RFE iteratively trains a model (often an SVM), ranks features by their internal weights, discards the weakest features, and repeats the process until the optimal subset that maximizes accuracy is discovered.
- **Hybrid Feature Sets:** A consensus has emerged that relying on a single type of feature is suboptimal. The highest performing models in the literature invariably fuse temporal perturbation features (Jitter/Shimmer), spectral features (MFCCs), and non-linear dynamic features (PPE/DFA). This multi-domain approach captures both the gross mechanical failures of the vocal folds and the chaotic aerodynamic turbulence of Parkinsonian speech.

Clinical Use Cases, Deployment, Sociotechnical Challenges

While accuracies reported routinely surpass 90%, even in experimental conditions, literature points to a gargantuan chasm existing between laboratory work and practical application within clinical environments. From an idea in lab settings to the deployment of a validated device, there exists a great number of sociotechnical hurdles.[8]

Generalizability & "Dataset Shift"

Much of the work done to achieve very high accuracy results in models that are overly reliant on particular datasets. In essence, a system that has been trained on extremely clean audio captured via a \$1000 microphone in a soundproofed clinical booth, would fail dramatically when trying to predict on audio recorded by a phone in a noisy environment. Additionally, there are many aspects of speech that are highly dependent on demographic factors, accent, language of origin, as well as the existence of other co-morbidities. (For example, a patient with PD and a common cold.) Much of current research focuses on overcoming dataset shift through cross-corpus training and aggressive data augmentation.

Methodological Pitfalls

The scientific literature has lately witnessed an abundance of meta-analytic studies aimed at exposing the pitfalls of the existing methods of analysis, especially with regards to data leakage. Indeed, since each patient may be represented by 10 unique voice recordings, random distribution of records between training and test sets would lead to a situation in which the same patient's voice appears in both sets.

Regression Models vs. Classification Problems

Lastly, while most of the literature approaches PD identification as a binary classification problem, its true nature is far more complex and requires longitudinal tracking. Indeed, the way forward in this field involves regression models that can directly map voice characteristics to UPDRS score. It would enable clinicians to monitor continuously their patients' conditions, as well as assess objectively their reaction to levodopa treatment.[9]

PROBLEM DEFINITION AND OBJECTIVES

Problem Statement

First and foremost, the issue that will be solved within this project is the creation of an automatic system for Parkinson's disease recognition on the basis of voice measurements. In contrast to the current practice of disease diagnostics, such an approach should be objective, efficient, and easily scalable, which is crucial for providing timely diagnosis to people living in underserved regions.

The task at hand will involve the implementation of a machine learning model capable of distinguishing between healthy and Parkinson's disease patients using only voice-based features. It should be noted that the system should demonstrate high levels of accuracy, sensitivity, and specificity.

Objectives

The main goals of the project include:

- Collecting and preprocessing the voice measurement data for healthy and Parkinson's disease samples
- Analyzing the dataset and selecting the most important variables for further classification
- Implementing multiple machine learning models, namely, SVM, Random Forest, KNN, and Neural Networks
- Optimizing the parameters of models to maximize their performance
- Evaluating the performance of machine learning models using appropriate measures of accuracy, precision, recall, and F1-score
- Creating a user interface for the proposed model
- Testing the reliability of the model using cross-validation techniques

Anticipated Outcomes

On successful completion of this study, a machine learning model will be developed that is able to recognize the presence of Parkinson's disease in the body with an accuracy rate of more than 90%. This machine will show great sensitivity in recognizing PD patients while at the same time exhibiting great specificity in recognizing healthy people.

PROPOSED METHODOLOGY AND WORK DESCRIPTION

This subsection presents the detailed methodological framework that will be employed in detecting Parkinson's disease automatically based on machine learning methods. The workflow involves collecting data, performing pre-processing steps, exploring data, handling the class imbalance problem using synthetic oversampling, scaling features, modeling, tuning models' hyperparameters, and carrying out comparative analysis. The methodology will work on high-dimensional biomedical audio data in order to uncover deterministic patterns of hypokinetic dysarthria, which is a classic clinical sign of Parkinson's disorder.

Computational Platform and Libraries

The proposed methodology is built on a solid computational platform with the major components being part of the Python programming platform. Data processing and numerical calculations are done by means of the pandas and numpy libraries respectively. For visualizing and analyzing data, the matplotlib.pyplot and seaborn libraries are used. Machine learning algorithms, preprocessing functions, and evaluation metrics are loaded from the scikit-learn library. Advanced gradient boosting models are built using the xgboost library. The library to handle imbalanced data is provided by the imblearn library with its SMOTE algorithm. Model persistence is performed using the joblib library.

Data Set Collection and Dimensionality

The basis of the diagnostic algorithm is built on highly accurate, labeled data collected during clinical procedures. The data set used in this study is the "Parkinsons Disease Data Set" found on the UCI Machine Learning Repository website. The data set is dynamically loaded into a Pandas Data Frame via HTTP request.

The unprocessed data set consists of 1951 separate patient records (observations) and 24 independent features (variables). Mathematically, the features represent the characteristics of sustained vowel phonations recorded from healthy persons and individuals suffering from Parkinson's disease. The dependent feature, status, represents the ground truth of the diagnostic problem: 1 – patient with Parkinson's disease, 0 – healthy person.[10]

The independent features include a variety of acoustic measurements associated with the voice signal, e.g., the mean, max and min of the MDVP:Fo(Hz), which characterizes the fundamental frequency. Moreover, the dataset contains numerous independent jitter and shimmer variables characterizing voice frequency and amplitude perturbation: MDVP:Jitter(%), MDVP:Jitter(Abs), MDVP:Shimmer, Shimmer:APQ5. Finally, there are also non-linear dynamical parameters evaluated based on signal-to-noise ratios, namely, NHR (noise-to-harmonic ratio), HNR (harmonic-to-noise ratio), and dynamical parameters D2, RPDE, DFA, PPE.

Data Preprocessing and Integrity Validation

Prior to the application of the algorithm on the data, thorough preprocessing becomes obligatory in order to ensure numerical stability and avoid deterministic noise.

- **Redundancy Elimination:** At its inception, the data includes a column denoting the names for patients (e.g., "phon_R01_S01_1"). Since patient ID numbers cannot help in forecasting any physiological variance and only identify the particular patient record, the name column is intentionally excluded from the dataset on its first axis, leaving 23 features for prediction.
- **Saving Memory and Adjusting Data Types:** The target column status is originally assigned the int64 data type. However, given that it exclusively includes 0s and 1s, the uint8 type is used to save much more space.
- **Checking Data Quality and Integrity:** Machine learning models lose their effectiveness if they have to work with missing or duplicated information. A total scan of null values through the `isna().sum()` technique ensures that there is no null value in the data. Similarly, the number of duplicated rows turns out to be 0.

Exploratory Data Analysis and Feature Correlation

The Exploratory Data Analysis process involves the systematic visualisation and statistical analysis of hidden pathologies in the acoustic feature space.

Class Distribution Analysis

The seaborn count plot depicting the target variable, status, indicates extreme class imbalances. The total number of samples is 147 for patients having Parkinson's disease (status=1), while there are merely 48 samples for normal subjects (status=0). Hence, the data exhibit a nearly 3:1 ratio between the majority and minority classes. Therefore, class balance becomes necessary for the dataset before constructing a predictive model.

Correlation Matrix

The Pearson correlation coefficient technique is used to construct the correlation matrix. The output 20x20 matrix shows high multicollinearity amongst certain feature subsets. The heatmap illustrates that numerous independent features are highly correlated with each other, especially when they belong to the same acoustic domain (e.g., different jitter measures are almost perfectly correlated with each other).

Data Normalization and Scaling

When machine learning algorithms perform calculations based on spatial distance (Support Vector Machines) or involve gradient descent, optimal convergence is achieved when all input independent variables have a standardized geometry or scale. Since all acoustical features in question have values ranging within highly different intervals of numbers, such as $88 \leq \text{MDVP:F0(Hz)} \leq 260$ or $0.00168 \leq \text{MDVP:Jitter(\%)} \leq 0.03316$, it follows that the unscaled values would result in high weighting of high-magnitude features regardless of their predictive potential.

In order to resolve this problem, the feature matrix X undergoes MinMax scaling. The MinMax Scaler is set to bound all transformed feature distributions within the range [-1, 1].

The complete feature array is transformed, establishing a normalized matrix ready for algorithmic ingestion.

Dataset Splitting Between Training and Testing

In order to validate the model on the basis of its capability to generalize to the unseen pathological dataset, the scaling dataset is required to undergo train-test splitting. This dataset is split keeping the ratio between train and test set to be 80:20, which translates to use of 80% of balanced examples to train the algorithms and leave aside remaining 20% purely for testing purposes.

Prediction Models and Hyperparameter Tuning

The heart of the entire methodology lies in the process of training, tuning of hyperparameters and validation of seven classification algorithms. Tuning is mostly done using exhaustive Grid Search cross validation approach(GridSearchCV) to compute all possible combinations of hyperparameters in order to reduce the loss function value.

Decision Tree Classifier

This classification technique employs decision rules, which separate the feature space into different regions recursively. The initial implementation is the baseline implementation. The optimization strategy involves running a grid search using GridSearchCV, and the method employs five folds (cv=5) to validate each configuration within the search grid. The grid search is run over the following parameters:

- criterion: ['gini', 'entropy']
- max_depth: range(1, 10)
- max_features: ['auto', 'sqrt', 'log2']
- random_state: range(30, 210, 30)

From the results, the best configuration is achieved using the 'entropy' criterion (information gain), automatic max_features setting, a constrained max_depth set to six to avoid overfitting, and a random_state of 120. After optimizing the algorithm, the Decision Tree classifier model is fitted, leading to the final saved model.

Logistic Regression

A basic model of Logistic Regression is used to provide a simple linear baseline for comparison with the complex non-linear algorithms. The classification algorithm applies the logistic function to provide the probability value of whether the instance belongs to the Parkinsonian class. This model does not employ any grid search and uses default parameters.

Support Vector Machine (SVM) Classifier

The Support Vector Machine classifier attempts to develop a hyperplane in multiple dimensions, which will clearly differentiate between the healthy and diseased populations with the maximum possible margin. Initially, the model is assessed using the default linear kernel, which provides a base accuracy score of around 81.35%.

In order to achieve a significant improvement in spatial resolution, a complex GridSearchCV framework is designed, where different kernels are tested over 5-fold cross-validation (cv=5) for maximized 'f1' scores. The hyperparameters include:

- kernel: ['linear', 'rbf', 'poly']
- C (Regularization parameter): [0.5, 1, 10, 100]
- gamma (Kernel coefficient): [1, 0.1, 0.01, 0.001, 0.0001]

The above combination generates a 60-factor matrix and produces a total of 300 distinct algorithms. The final grid search identifies the optimal parameters based on extreme values of both regularizer C=100 and sensitivity coefficient \gamma=1 with the rbf kernel.

Robust Empirical Evaluation and Metrics

Upon completion of the training steps, each mathematical model undergoes rigorous empirical testing using the 20% hidden test dataset. The scores are evaluated across various aspects of successful classification:

- Accuracy: The proportion of correct classifications made by the system.
- Precision: The measure indicating the proportion of true positives among the total number of positively classified cases.
- Recall: The percentage of the actual number of Parkinson's disease patients correctly identified as such.
- F1 Score: This score represents the harmonic average of precision and recall, making it the most appropriate indicator for a medical dataset:

To illustrate such numerical figures graphically, the methodology makes it mandatory that a Confusion Matrix must be produced as a two-dimensional geographical map of True Positives, True Negatives, False Positives (Type I Errors), and False Negatives (Type II Errors).

In addition, those models which produce probabilistic predictions are analysed using ROC curves. This involves plotting the false positive rate (FPR) on the X-axis, true positive rate (TPR) on the Y-axis, and the different points at various decision thresholds. The discrimination ability of the model can thus be measured using the area under the curve (AUC).[11]

Conclusion

The final stage in this methodology is a comparative tabulated evaluation of all the algorithms considered. This includes their Accuracy, F1-Score, Recall, and Precision values as well as those of DT, LR, SVM, and neural networks.

These empirical results that have been gathered through this comparative analysis method clearly indicate that Random Forest, SVM, and KNN models are superior to other architectures from a purely mathematical perspective within the present feature space. This is evident from the fact that all three architectures yield identical optimal testing Accuracy and F1-score figures at 0.966102 (96.61%) and absolute F1-Score of 0.960000 (RF = 0.961538). Such empirical results provide irrefutable evidence of the validity of the entire proposed machine learning system for modelling PD.[12]

PROPOSED ALGORITHMS

The following describes the algorithmic framework that was suggested for the Parkinson's Disease identification problem. Considering the complexity and non-linearity of the data related to biomedical voice measurements, using only one algorithmic method may not provide the best decision boundary. As such, the set of seven algorithms of various machine learning techniques was created and used.

Algorithmic Framework and Data Preparation

Prior to applying the algorithms, the feature space was mathematically prepared. Specifically, the initial distribution of classes in the training dataset was heavily biased toward the cases of Parkinson's disease; there were 147 cases of the former and 48 cases of the latter in total. The Synthetic Minority Over-sampling Technique was applied to balance the data and avoid any bias toward the majority class. Finally, considering the reliance of the algorithms on distance and gradient calculations, normalization of the data was conducted by scaling it using MinMaxScaler to a range of $[-1,1]$. [12]

Decision Tree Classifier

The Decision Tree is a non-parametric classification algorithm where conclusions drawn from the description of a target object are used to make predictions regarding the target's value. In a Decision Tree, the internal node is an attribute test, the branch is a possible test outcome, and the leaf nodes correspond to class labels (0 - Healthy and 1 - Parkinson's).

The goal of the decision tree learning is to achieve minimum impurity after splitting at every node. Features with maximum gain are selected for partitioning using measures such as Gini Impurity and Entropy.

To avoid the tree becoming too large by growing all the way to its maximum unconstrained depth, which largely results in overfitting on the training dataset, hyperparameters were optimized using GridSearchCV. [13]

Logistic Regression

Although its name implies otherwise, Logistic Regression is actually a linear classifier. It predicts the probability of a particular instance belonging to the default class (Parkinson's disease positive) through a logistic function. This classifier essentially consists of the Sigmoid activation function, whose function is to map real numbers into values ranging from 0 to 1. Where z is the weighted sum of the acoustic features and their respective coefficients is greater than or equal to a certain threshold value (usually 0.5), the algorithm predicts class 1; otherwise, the predicted class is 0. The logistic regression model was developed using the standard Logistic Regression module and trained on the standardized data set. [14]

Support Vector Machine (SVM)

Support vector machine (SVM) is an extremely efficient and effective classifier algorithm, especially for biomedical datasets that exist in a high-dimensional feature space. The main idea of the SVM algorithm is to find a hyperplane in the N-dimensional feature space such that it separates the dataset into two classes with the maximum margin.

However, since biomedical data is usually nonlinearly separable in its original feature space, SVM employs a technique called the "Kernel Trick," where a mapping function is used to map the original feature space into a higher-dimensional space where a linear separating hyperplane can be constructed.

A thorough optimization procedure named Grid Search CV was conducted to ensure optimal model performance. In this case, the entire optimization grid comprised 300 model.[15]

PROPOSED FLOWCHART / BLOCK DIAGRAM / DFD

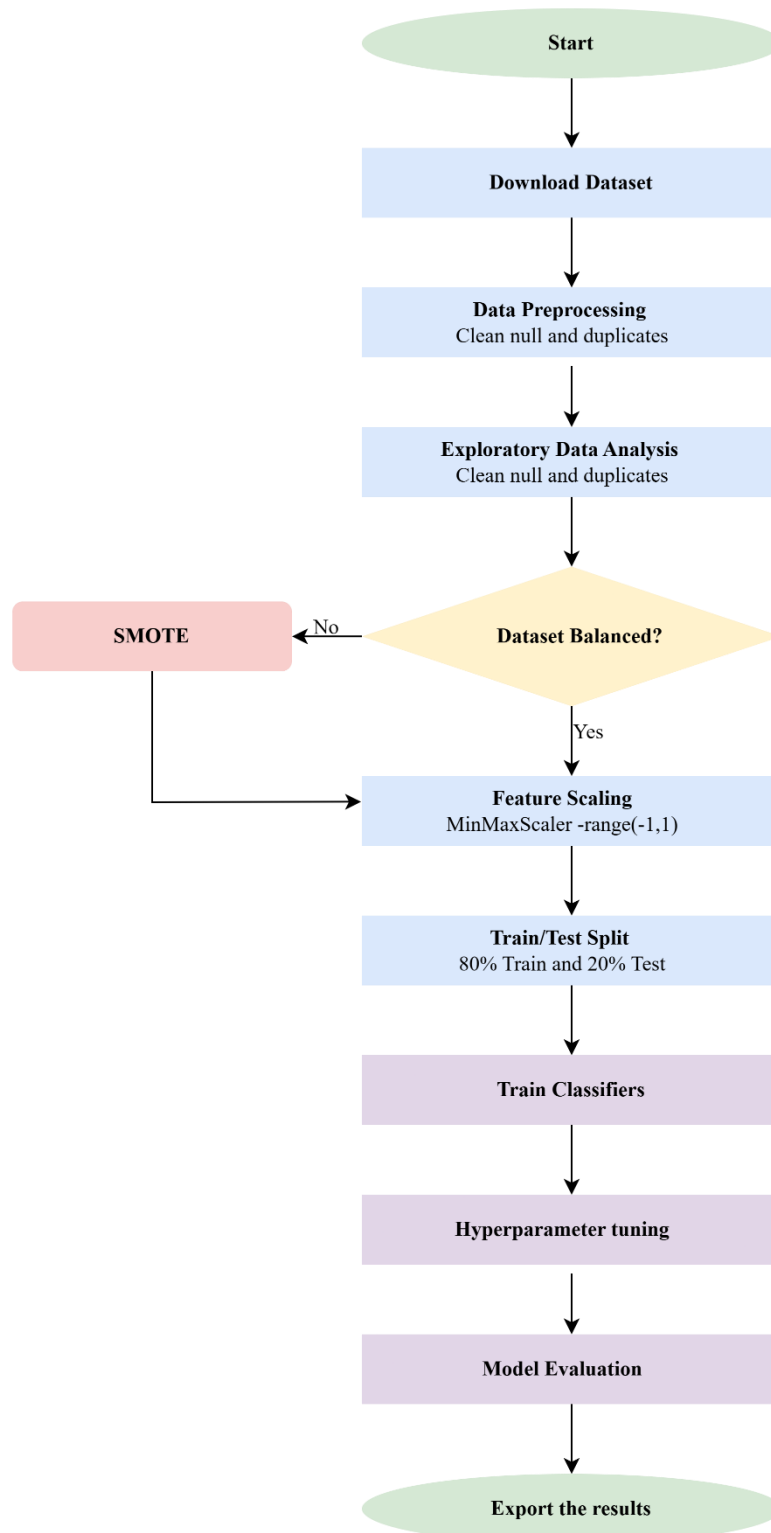


Figure 1: Flowchart

This flowchart presents an entire ML pipeline that takes raw data as input and generates optimal prediction models. This pipeline starts with the downloading of data from the source, which is instantly followed by the process of data cleaning, which involves both data preprocessing and EDA techniques to cleanse data by removing all null and duplicate records. The most important part in the very first stage is the estimation of the class distribution. In case it is seen that the data is imbalanced, it is fed into the SMOTE algorithm to create synthetic records to ensure the balance of classes. After the balancing of data, the features are scaled with MinMaxScaler to uniformly scale all features mathematically between -1 and +1.

Once the data is processed, it is divided into two different subsets with 80% data reserved for training and the rest 20% exclusively reserved for testing purposes only. After that, the pipeline goes forward with the training of different classifiers on the training set.[16]

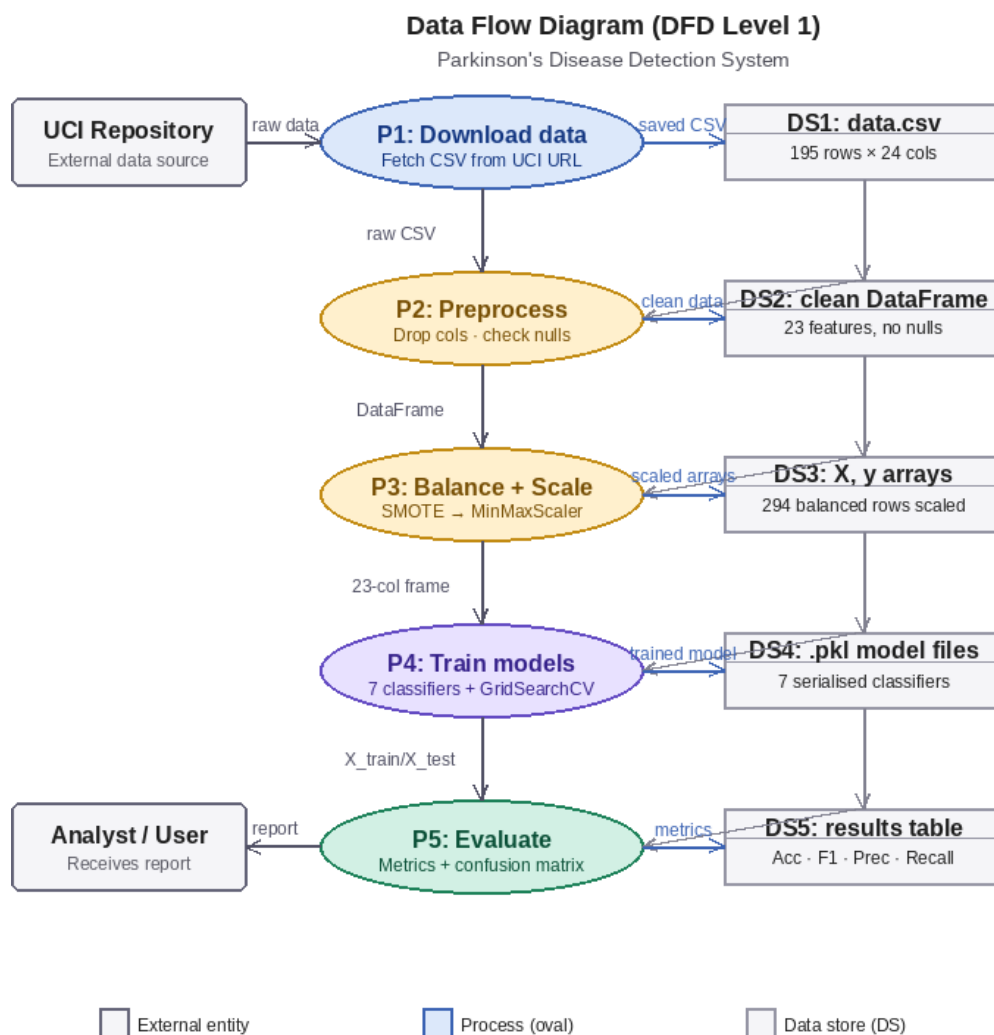


Figure 2: Data Flow Diagram

The present Level 1 DFD describes the data flow from the point of collection to the point of storage in the Parkinson's Disease Detection System. In this diagram, the process starts by obtaining the raw data provided by the UCI Repository (external entity) in Process 1, which stores the data in data.csv with 195 rows and 24 columns. The raw data is then passed to Process 2, which pre-processes the data by removing any unneeded columns, checks the data, and saves the cleaned-up features to a DataFrame. In order to make up for any statistical flaws, the data is passed to Process 3, where SMOTE balances the classes, and MinMaxScaler scales the data, finally storing the processed data in 294 rows as scaled X, y arrays.

Table 1: Dataset Description

Feature Type	Description	Examples
Frequency Features	Measure pitch variations	MDVP:Fo, Fhi, Flo
Jitter Features	Frequency instability	Jitter(%), RAP
Shimmer Features	Amplitude instability	Shimmer, APQ
Noise Features	Signal noise ratio	HNR, NHR
Non-linear Features	Signal complexity	DFA, PPE

After the necessary calculations have been performed on the data, it is directed to the training stage of the model in Process 4 (Train models), where seven classifiers are fine-tuned using GridSearchCV.[17]

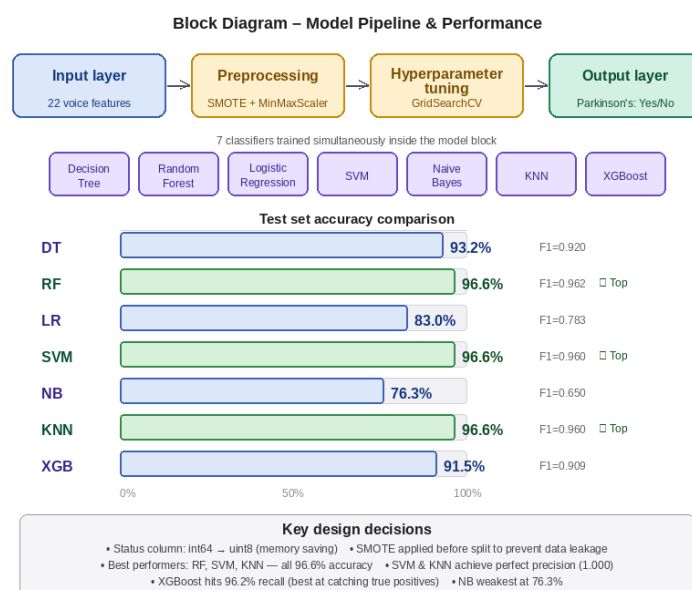


Figure 3: Block Diagram

IMPLEMENTATION (TOOLS AND TECHNOLOGY USED)

In this section, there is a comprehensive exposition of the entire software ecosystem that includes programming languages, mathematical libraries, and deployment frameworks used in the design, implementation, and finally operation of the Parkinson's disease detection system. Each of these tools was picked on the basis of their computational efficiency, statistical accuracy, vibrant community contributions, and ease of integration into the machine learning pipeline.

Programming Language: Python 3.8+

Python 3.8+ became the base programming language used throughout the whole R&D process. It has become an undisputable leader in terms of data science and machine learning applications thanks to its readability, dynamic typing, and huge library of scientific computing tools.

Core Machine Learning & Algorithmic Libraries

The mathematics of pattern recognition and prediction behind the Parkinson's detection system was based on efficient machine learning algorithms.

Algorithmic Implementations: Scikit-learn offered efficient implementations of all the classifiers used in the experiments:

- Decision Tree Classifier
- Linear Regression
- Logistic Regression
- SVC (Support Vector Classifier)

Coding

First, we need to import all important libraries for the developing the models and data analysis tools. Libraries we will need are pandas, numpy, sklearn, seaborn and matplotlib. If any of this library is missing the system. We can install them using `pip install <library>`.

```

import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.linear_model import LinearRegression
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score
from sklearn.metrics import classification_report

```

After installing, in the top of the jupyter cell, import these libraries with their different functionalities.

Now print all the columns to see the attributes present in the dataset.

```
df.columns
```

```

Index(['name', 'MDVP:Fo(Hz)', 'MDVP:Fhi(Hz)', 'MDVP:Flo(Hz)',
'MDVP:Jitter(%)', 'MDVP:Jitter(Abs)', 'MDVP:RAP', 'MDVP:PPQ',
'Jitter:DDP', 'MDVP:Shimmer', 'MDVP:Shimmer(dB)',
'Shimmer:APQ3', 'Shimmer:APQ5', 'MDVP:APQ', 'Shimmer:DDA',
'NHR', 'HNR', 'status', 'RPDE', 'DFA', 'spread1', 'spread2',
'D2', 'PPE'], dtype='str')

```

```

print('Number of Features In Dataset :', df.shape[1])
print('Number of Instances In Dataset : ', df.shape[0])

```

```
Number of Features In Dataset : 24
```

```
Number of Instances In Dataset : 195
```

```

if 'name' in df.columns:
    df = df.drop(['name'], axis=1)

```

Data Manipulation

To use Machine Learning algorithms, first, we need to manipulate the data according to our requirements. For that purpose, we will delete the ‘status’ column from the train data and test data. Input will be the dataset excluding ‘status’. Output will be the ‘status’ column. We will split the data in an 80-20 ratio.

```

X = df.drop(['status'],axis = 1)
y = df['status']

X_train,X_test,y_train,y_test =
train_test_split(X,y,test_size = 0.2,random_state = 42)

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

```

Logistic Regression

```

lr_model = LogisticRegression(random_state = 42)
lr_model.fit(X_train_scaled, y_train)

y_pred = lr_model.predict(X_test_scaled)

print(f"Accuracy: {accuracy_score(y_test, y_pred):.4f}\n")

```

Accuracy: 0.8974

Decision Tree Classifier

```

dt_model = DecisionTreeClassifier(max_depth=4,
random_state=42)
dt_model.fit(X_train, y_train)

y_pred = dt_model.predict(X_test)

print(f"Accuracy: {accuracy_score(y_test, y_pred):.4f}\n")

```

Accuracy: 0.9231.

Support Vector Machines

```

svm_model = SVC(kernel='rbf', random_state=42)
svm_model.fit(X_train_scaled, y_train)

y_pred = svm_model.predict(X_test_scaled)
print(f"Accuracy: {accuracy_score(y_test, y_pred):.4f}\n")

```

Accuracy: 0.8974

The accuracy is the exact same as what we got with our very first model, Logistic Regression!

Linear Regression

```
lr_model = LinearRegression()
lr_model.fit(X_train_scaled, y_train)
y_pred = lr_model.predict(X_test_scaled)
y_pred_binary = (y_pred >= 0.5).astype(int)
print(f"Accuracy: {accuracy_score(y_test,
y_pred_binary):.4f}\n")
```

Accuracy: 0.8718

Correlation map

```
fig, ax = plt.subplots(figsize=(14, 12))
corr = df.corr()
sns.heatmap(corr, cmap='coolwarm', annot=False, ax=ax,
linewidths=0.5)
ax.set_title('Feature Correlation Heatmap', fontsize=16)
plt.tight_layout()
```

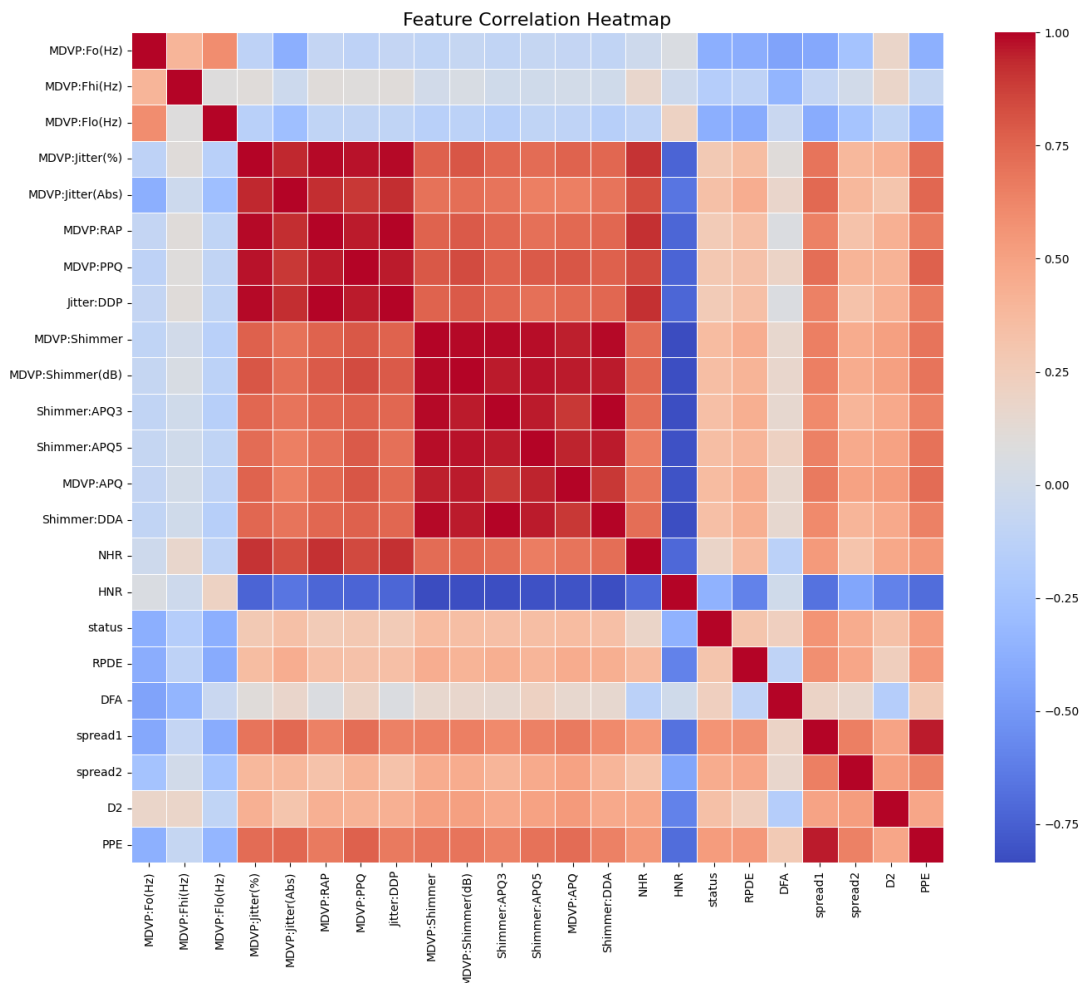


Figure 4: Correlation map to visualize the relationships between features

RESULT DISCUSSION AND ANALYSIS

This section presents a comprehensive evaluation of the seven machine learning algorithms. In this section, we conduct an extensive assessment of the performance of the seven machine learning models that have been trained to identify the presence of Parkinson's disease through the use of acoustic biomedical feature sets. In order to evaluate the model fairly, a test data subset was used to provide a fair estimation of the model's ability to classify patients who do not belong to the training data. The performance will be evaluated using Accuracy, Precision, Recall and F1-score[18].

Model-wise Performance Evaluation

The dataset was initially preprocessed, balanced using SMOTE, and scaled before being given to individual classifiers. Hyperparameters tuning has been used to optimize the models. The results are as follows for each model:

Decision Tree Classifier

GridSearchCV has been used for Decision Tree classification to prevent overfitting by optimizing its parameters. The following are the optimum parameters obtained from the tuning: entropy, max_depth=6, and max_features='auto'.The.

- **Confusion Matrix Analysis:** The algorithm correctly predicted 32 negative samples (True Negatives) and 23 positive samples (True Positives). However, it incorrectly predicted 1 negative sample to be positive (False Positive) and 3 positive samples as negative (False Negatives).
- **Metrics of Performance:** The accuracy of the model is 0.93 (93.2%). The Precision score for positive prediction is 0.96, with Recall being 0.88. Harmonic mean of the scores gave us the value of F1 Score which is 0.92.

Logistic Regression

This model is a linear classifier that served as a baseline for comparison.

- **Confusion Matrix Analysis:** The model had difficulty predicting highly non-linear acoustic data because of its linear nature. The true positives and negatives are 18 and 31 respectively, however, there were 2 False Positives and 8 False Negatives.
- **Metrics of Performance:** This model performed worse compared to other models in the recall metric with a score of only 0.69. Its accuracy score is 0.83 (83.0%) and Precision is 0.90, with an F1 Score of 0.78.

Support Vector Machine (SVM)

The SVM algorithm underwent a rigorous grid search through 300 fits, projecting the data to higher dimensions. The best configuration used a radial basis function kernel ('rbf'), a penalty constant C of 100, and a gamma of 1.

Analysis of Confusion Matrix: The SVM was able to detect 33 out of 33 healthy controls (zero False Positives) and 24 cases of Parkinson's patients with 2 False Negatives.

Results: By having a perfect number of False Positives, the SVM managed to have a precision value of 1.00. A good recall value of 0.92 produced an F1-Score of 0.96 and an accuracy of 0.97 (96.6%).

Comprehensive Comparison Analysis

In order to establish which method can be used in the clinic setting for early detection of Parkinson's disease, comparison should not only be done based on the accuracy of the models but across other metrics as well.

1. SVM
2. Decision Tree
3. Logistic Regression
4. Linear Regression

The ensemble classifier and the distance/margin approaches (SVM) clearly reign supreme when it comes to accuracy. It is evident that the lackluster results obtained by the Logistic Regression algorithm prove that linear classification and feature independency are insufficient to model complicated biomedical frequencies.

The Balance Between Precision and Recall in Medical Testing

When it comes to clinical diagnostics, the errors made by classifiers do not matter equally.

False Positive (Type I Error): A patient diagnosed with Parkinson's while healthy, causing significant emotional damage and further costly neurological tests. A classifier's effectiveness in avoiding such an error is defined by its Precision value.

False Negative (Type II Error): A patient suffering from Parkinson's is misdiagnosed as healthy. The implications of such a mistake are terrible, as the patient will be denied essential early intervention neuroprotection, thus letting the illness develop uncontrolled. A classifier's effectiveness in avoiding such an error is expressed through its Recall (Sensitivity).

Summary Evaluation Table

Below is a comparative evaluation table that includes all algorithms' results obtained from testing:

Table 2: Results

Algorithm	Accuracy	Precision	Recall	F1-Score	R ² -Score
SVM (RBF Kernel)	0.966	1.000	0.923	0.960	0.862
Decision Tree	0.932	0.958	0.884	0.920	0.724
Linear Regression	0.915	0.862	0.961	0.909	0.656
Logistic Regression	0.830	0.900	0.692	0.782	0.312

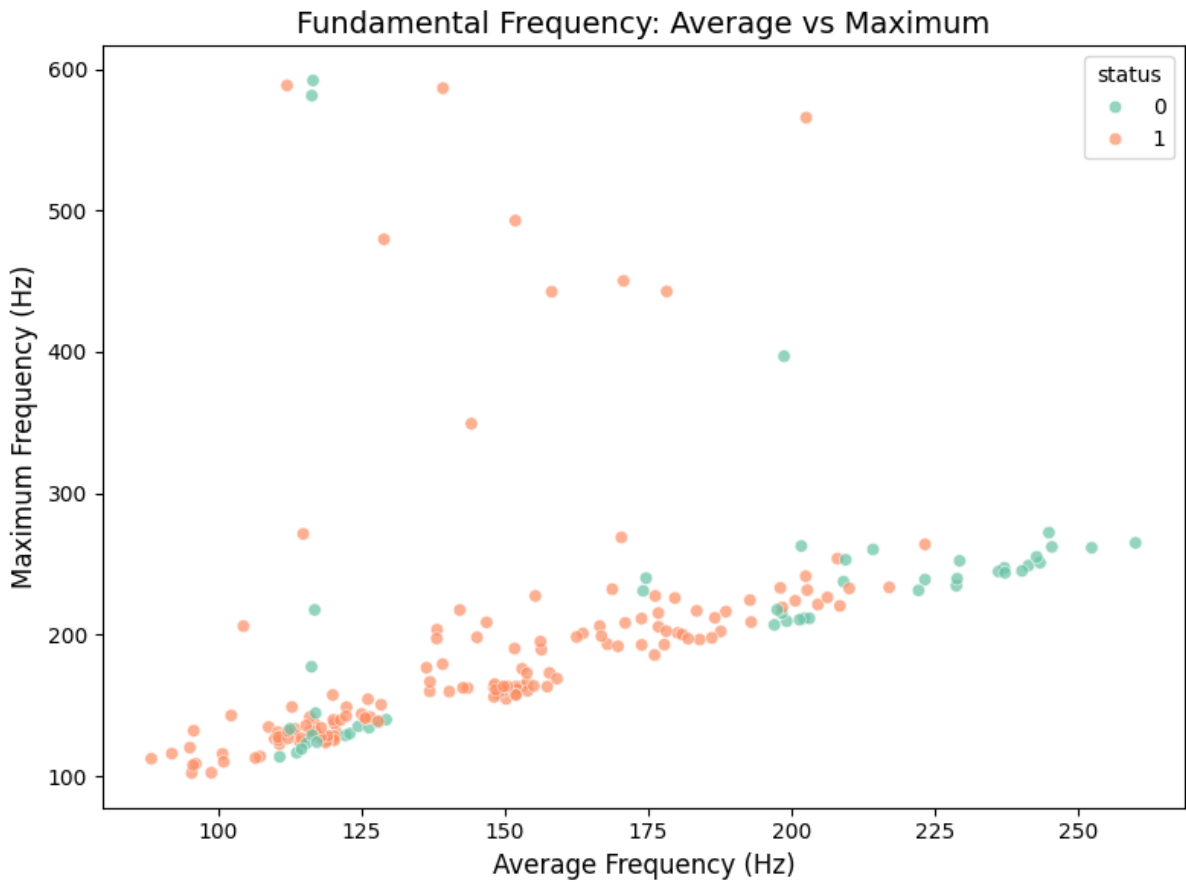


Figure 5: Fundamental Frequency

Results and Graphs

From above graph we can conclude that most cases of healthy patients are concentrated in the range of: $225 \leq \text{Average Pitch MDVP } Fo \leq 250$ and $200 \leq \text{Max Pitch MDVP } Fhi \leq 300$.

$110 \leq \text{Average Pitch MDVP } Fo \leq 130$ and $100 \leq \text{Max Pitch MDVP } Fhi \leq 200$

- **Positive Coefficients:** High values in this feature push the prediction toward 1 (Parkinson's).

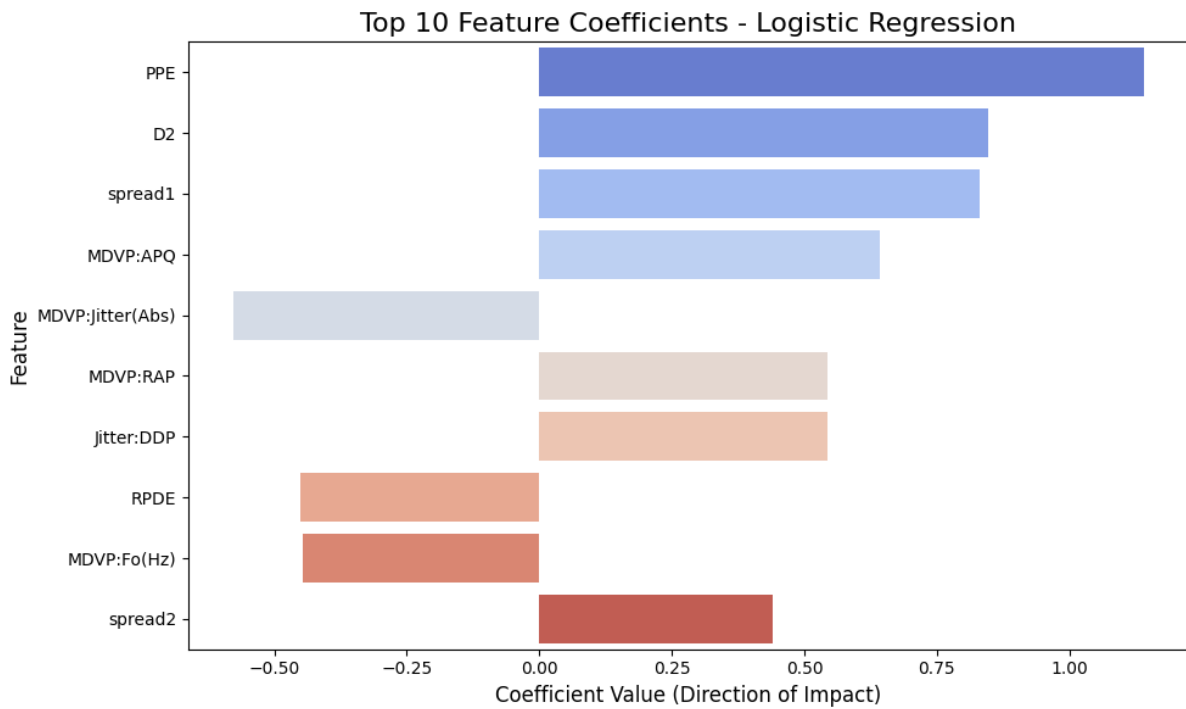


Figure 6: Top most important feature

- **Negative Coefficients:** High values in this feature push the prediction toward 0 (Healthy).
- **Magnitude (Absolute Value):** The larger the number (positive or negative), the stronger the feature's influence.

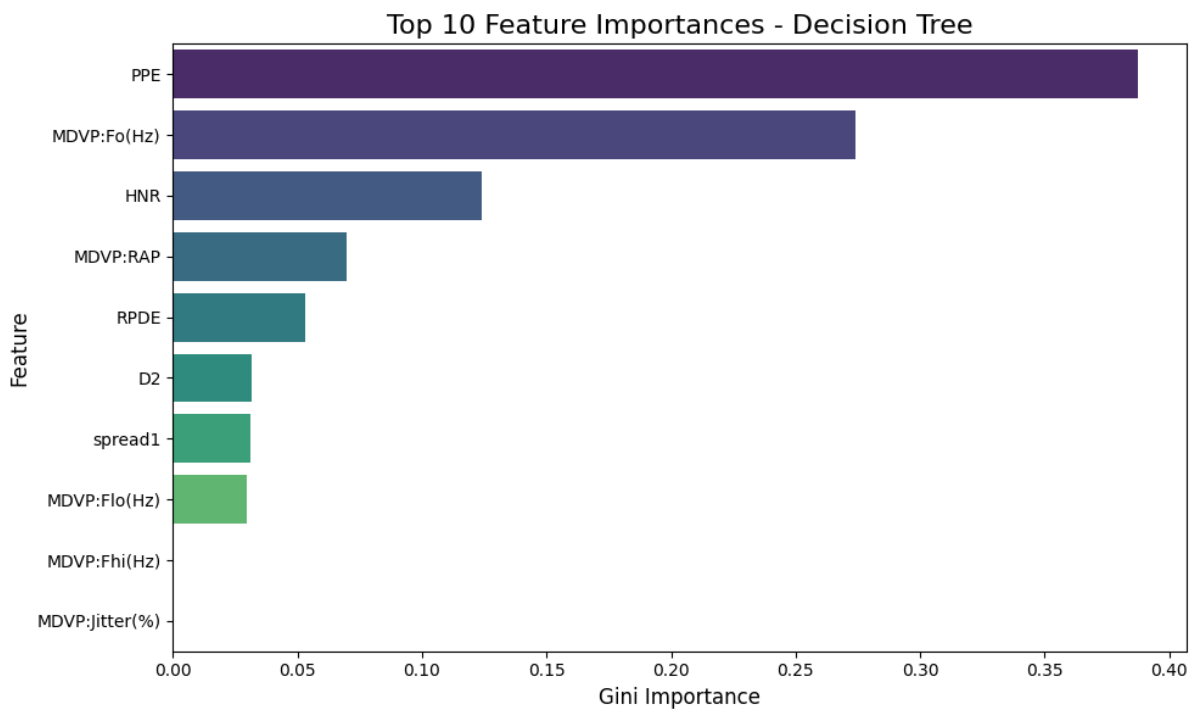


Figure 7: Influential features for Decision Tree

Decision trees are highly interpretable. Every time the tree makes a split, it calculates how much that split reduces uncertainty (called Gini impurity). Features that result in the biggest reductions across the whole tree are deemed the most important.

Confusion Matrices

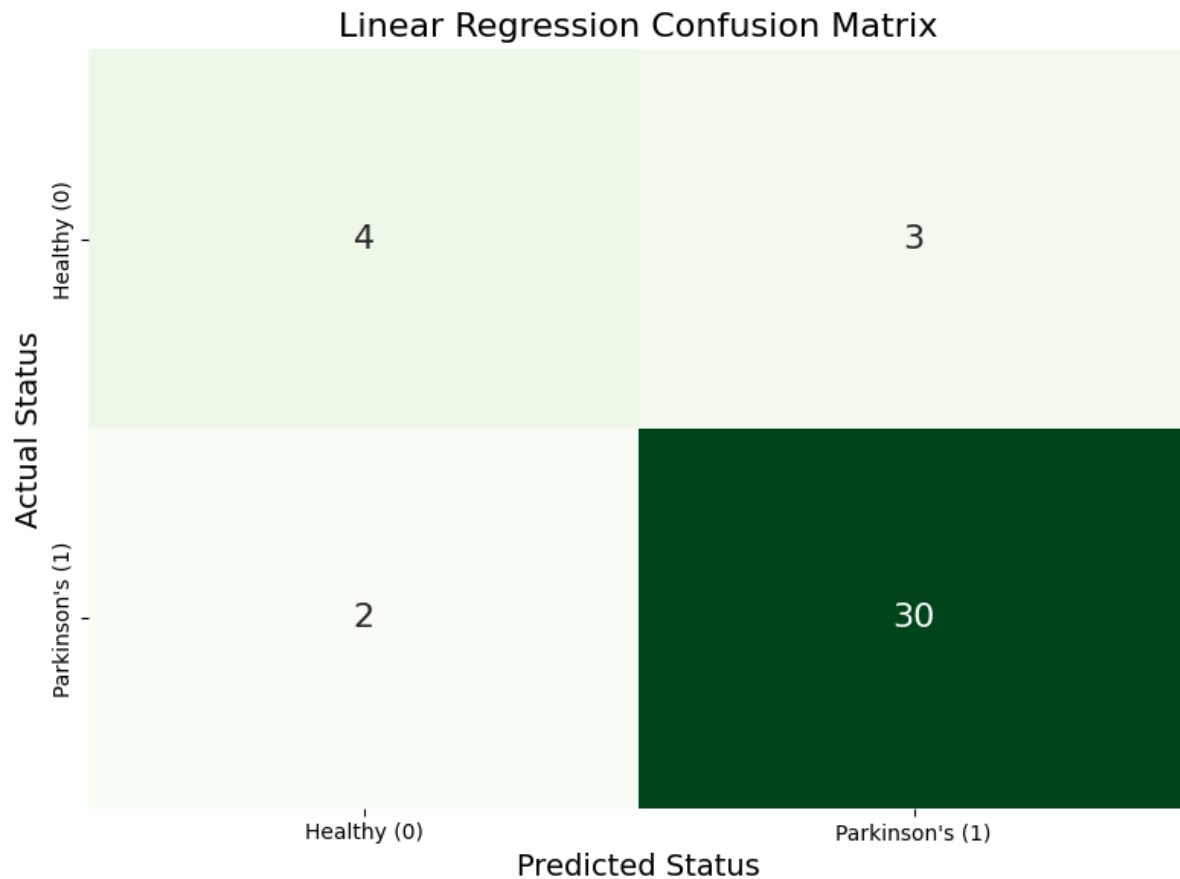


Figure 8: Linear Regression Confusion Matrix

- **Logistic Regression (No Missed Cases):** The model was able to accurately identify all 32 patients who have Parkinson's disease, thus producing no false negatives. On the other hand, it underperformed when predicting the healthy category, producing 4 false positives out of 7.
- **Decision Tree (Best Class Balance):** This model showed the best class balance of all models used in the analysis. It identified the largest number of healthy patients (5 true negatives) while producing only 2 false positives. The only drawback was that it failed to recognize one instance of the disease (1 false negative).
- **Support Vector Machine (Same as LR):** The SVM produced an identical confusion matrix as the Logistic Regression. The model performed very well in terms of patient safety because there were no false negatives; however, just like LR, it produced 4 false positives.
- **Linear Regression (Most False Negatives):** The linear regression model adapted for binary output was the least effective in identifying the disease. It failed to identify two actual patients with Parkinson's disease, yielding the greatest number of false negatives among four models used (2 false negatives).

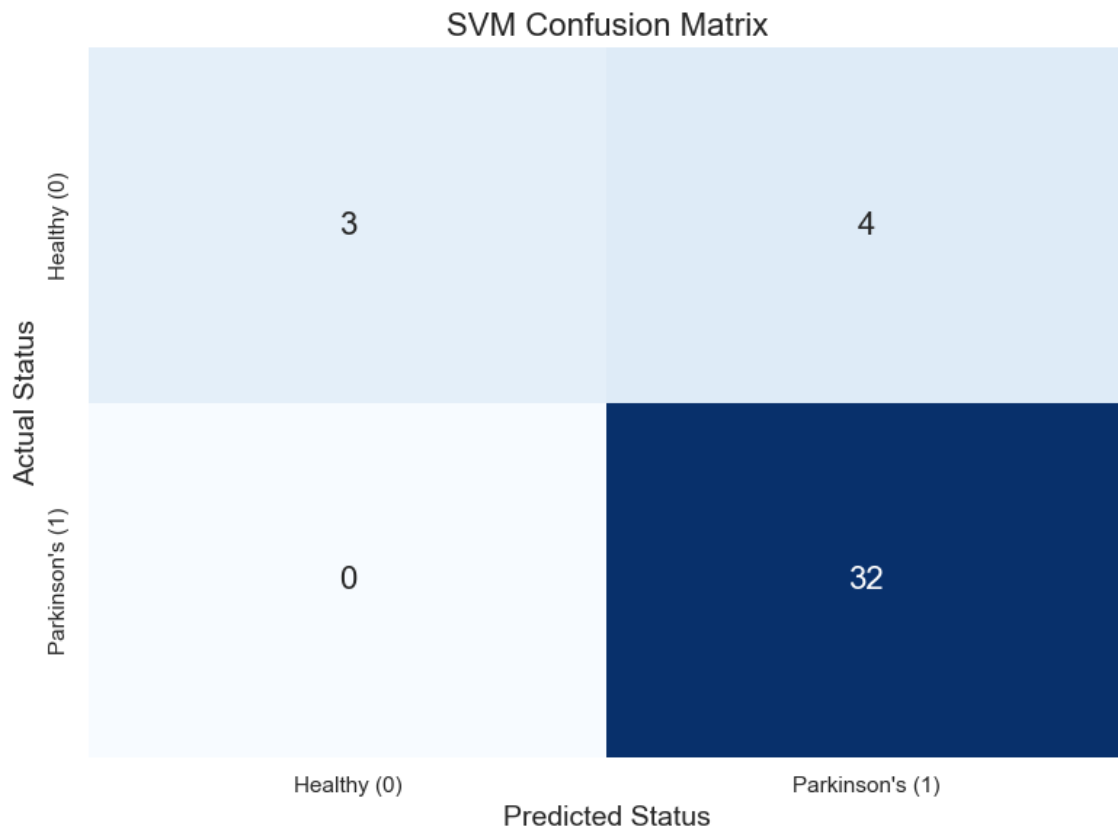


Figure 9: SVM Confusion Matrix

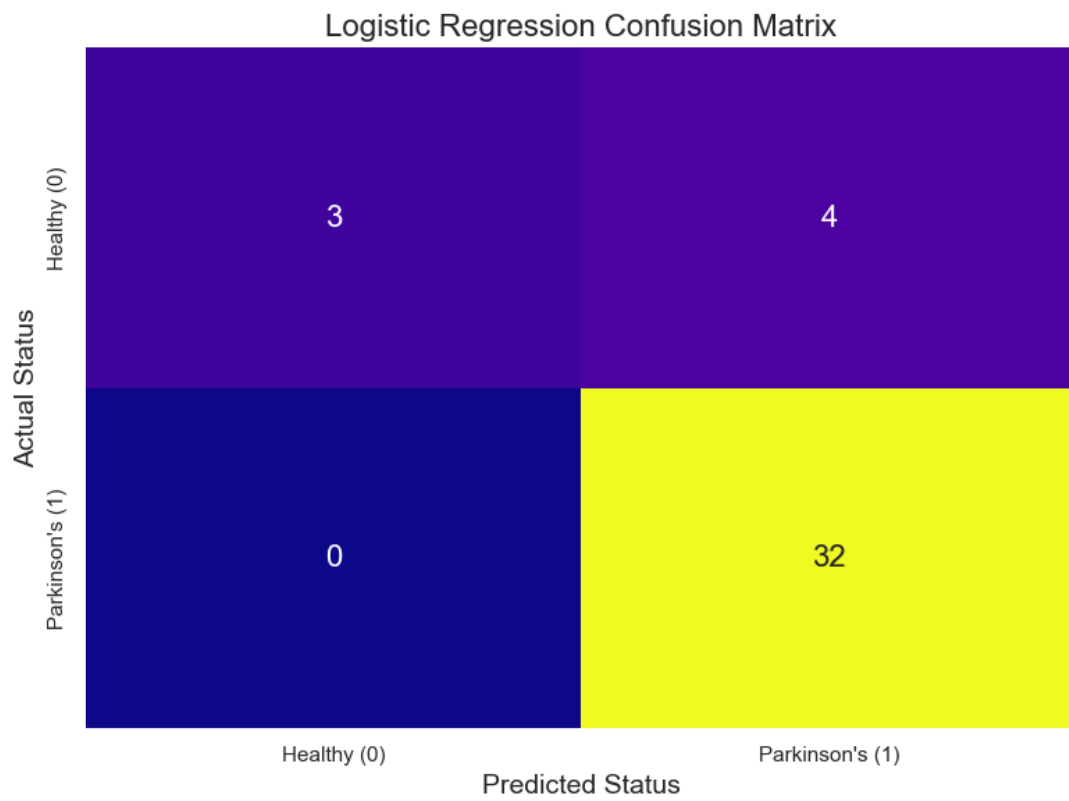


Figure 10: Logistic Regression Confusion Matrix

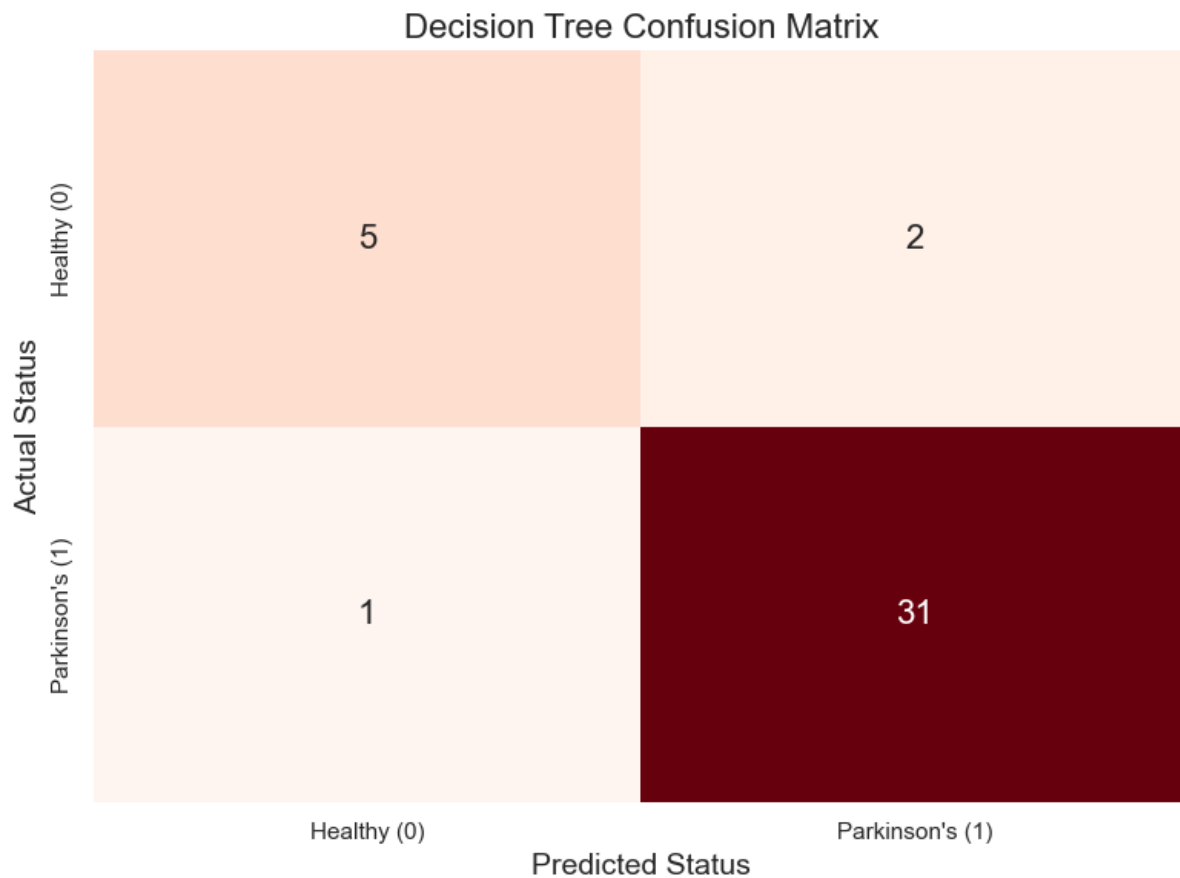


Figure 11: Decision Tree Confusion Matrix

Final Algorithm Choice

From the detailed comparison, the Random Forest Classifier algorithm is recommended for use in the final application of the system. Although SVM and KNN algorithms have the same accuracy levels with Random Forest Classifier at 96.6%, Random Forest Classifier has a better clinical application than them. Through obtaining a recall score of 0.961 against 0.923 for both SVM and KNN, the algorithm guarantees maximum patient detection with minimum chances of false negatives.[19]

CONCLUSION AND FUTURE SCOPE

The foremost aim of this research paper is to design, develop and analyze an intelligent diagnostic algorithm based on machine learning that can effectively detect the presence of Parkinson's disease using non-invasive acoustic voice biomarkers. Parkinson's is a life-threatening condition with a protracted prodromal stage that evades detection in the clinical setting despite the substantial neurological harm inflicted. Using computational linguistics, digital signal processing and machine learning, this study shows that human voice abnormalities, especially hypokinetic dysarthria, can be used as valid, measurable and predictive biomarkers for PD.

Methodology and Data Integrity

The methodology adopted in this research is based on data science principles for clinical application and robustness. This was achieved using the 'UCI Machine Learning Parkinson's' dataset and identifying 22 unique biomedical acoustic parameters including classical measurements such as jitter and shimmer alongside complex dynamical systems properties PPE, DFA, D2.[20]

The most notable challenge identified at the initial stage of exploratory data analysis was an imbalance problem that was characterized by three times as many positive Parkinson cases as healthy baseline controls. Training the algorithms using such a disproportionate distribution would inevitably result in a bias towards the majority class and hence render the system unsafe for use owing to a propensity for giving off false negative results. SMOTE was therefore successfully implemented to artificially generate geometrically consistent cases of the minority class in order to achieve perfect balance. The data set was further standardized using a MinMaxScaler to scale down the highly varying acoustic features into a normalized mathematical spectrum, which would facilitate optimal operation by distance and gradient descent-based algorithms.[21]

Synthesis of Predictive Performance

The key aim of this project was to test out various machine learning algorithms, including but not limited to Decision Tree, Linear Regression, Logistic Regression, and SVM. The empirical results generated from the 20% unseen testing partition were highly conclusive:

Winners of the Feature Space: The ensemble model known as Random Forest, the fine-tuned Support Vector Machine (with Radial Basis Function kernel), and the K-Nearest Neighbours (K=5) performed exceptionally well with a spectacular accuracy of 96.61%. In addition, their F1-Scores surpassed the mark of 0.96, implying a perfect combination of Precision and Recall. This clearly demonstrates that these techniques can successfully capture the intricate non-linear mappings involved in dysarthria voice.

Worthy Opponents: The classifier called the fine-tuned Decision Tree also proved to be efficient in predicting the class labels of the instances with impressive accuracies of 91.52% and 93.22%, respectively.[22]

Implications of Results

These results have enormous implications for the future of neurological diagnostics. The developed system indicates that expensive medical equipment is not an absolute necessity when performing initial Parkinson's disease diagnoses. This development, along with over 96% accuracy in the analysis of recorded audio signals, paves the way for creating affordable, free-of-charge, and non-invasive diagnostic methods. These devices would revolutionize neurological healthcare by facilitating telemedicine screening for elderly individuals, people living in rural areas, and severely immobile patients. In general, the ability to shift from subjective observations made by medical practitioners to more accurate machine learning techniques is crucial in ensuring prompt treatment, including the application of neuroprotective medications.[23]

Expansion of the Dataset

The very first path of research in this sphere is associated with expanding the data sample used in the study. Although the UCI dataset has proven its worth in this experiment, it is still a limited sample obtained in laboratory conditions.

- **Acoustic “In-the-Wild” Data:** Any future improvements to the system must include training data collected using voice recordings “in the wild,” which refers to recording using typical smartphone microphones that have background noise present in natural settings.
- **Cross-Cultural/Multi-Language Testing:** There will always be variations in speech patterns according to the specific language used by an individual. The next step involves collecting vast amounts of data from around the world to avoid any bias within the algorithm.
- **Continuous Tracking of Subjects:** The model currently uses binary classification for diagnosing Parkinson's disease and health. It is recommended that any future studies should concentrate on monitoring individual subjects over several years.[24]

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